Student Handout: Euler Method Programming Structure

Example Problem: Radioactive Decay

- ODE: dN/dt = -kN
- Initial Condition: N(0) = N0
- We will solve this using the Euler method in Python.

1. Import Required Packages

- import numpy as np
- import matplotlib.pyplot as plt
- These packages allow numerical operations and visualization.

2. Define Constants and Initial Conditions

- N0 = 1000 # Initial number of nuclei
- k = 0.1 # Decay constant
- dt = 0.1 # Time step (s)
- These define your problem setup.

3. Initialize Arrays

- N = [N0]
- t = [0]
- Use lists to store the history of values.

4. Choose Loop Type

- Use a while-loop if you want to stop when N is very small.
- Use a for-loop for a fixed number of time steps.

5. Euler Update Rule

- dNdt = -k * N[-1]
- N_next = N[-1] + dNdt * dt
- Apply the numerical approximation to update the value.

6. Append to Arrays

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- N.append(N_next)

- t.append(t[-1] + dt)

- Track values as they evolve over time.

7. Plotting Results

- Use matplotlib to visualize N versus time.

- Label your axes and give a meaningful title.

Summary: Code Structure

- 1. Import packages

- 2. Declare constants and ICs

- 3. Initialize arrays

- 4. Choose loop type

- 5. Apply Euler update rule

- 6. Update arrays

- 7. Plot results